SOLVING THE RANGE CHALLENGE In IoT

Are walls and distance killing your smart network? See how Zigbee's new sub-GHz Suzi extension shatters range limits.



The Connectivity Standards Alliance announces Zigbee 4.0, and Suzi, the new brand for Zigbee's Sub-GHz feature. These are two major milestones in advancing the foundation of secure, interoperable, and scalable IoT connectivity.

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How far can wireless devices communicate before range fails? While small-home setups are straightforward, large premises, manufacturing plants, and outdoor sites present severe connection challenges due to distance and physical barriers. Traditionally, developers turn to Zigbee, a low-power, 2.4GHz mesh networking protocol built specifically for the Internet of Things (IoT) devices. However, many deployments require deeper penetration and longer reach than the standard 2.4GHz band can provide.

Enter Suzi, an open, non-proprietary sub-GHz extension of the Zigbee ecosystem operating in lower-frequency bands (800MHz in Europe and 900MHz in North America). By shifting to these lower frequencies, Suzi delivers massive range boosts and superior wall penetration. Crucially, it builds directly on top of existing Zigbee network-layer routing. Consequently, it keeps Zigbee's hallmark low power consumption and self-healing mesh topology intact, while simultaneously giving manufacturers complete freedom in how they implement the hardware.

Why sub-GHz matters

Operating in the sub-GHz spectrum bypasses the crowded 2.4GHz band, which is heavily congested by Wi-Fi, Bluetooth, and standard Zigbee traffic. This frequency shift significantly improves link reliability and extends line-of-sight range to 1km or more, according to the Connectivity Standards Alliance (CSA). Final performance, however, remains dependent on transmission power, antenna design, and environmental factors.

For battery-powered IoT hardware deployed across factories, security systems, and outdoor infrastructure, this extended reach drastically reduces battery maintenance cycles. Furthermore, Suzi retains Zigbee's native self-healing mesh architecture, allowing devices to maintain connectivity through alternate network paths without requiring direct communication with a central gateway.

How Suzi works with existing Zigbee networks

Suzi extends the existing Zigbee ecosystem rather than replacing it.

While the underlying network-layer architecture remains identical, Suzi introduces standalone or hybrid sub-GHz operation alongside legacy 2.4GHz deployments. Dual-radio bridge devices enable seamless inter-network communication, allowing developers to expand existing installations incrementally. A hybrid setup can retain 2.4GHz nodes for short-range indoor coverage while deploying Suzi nodes only where extended range or superior barrier penetration is required.

Additionally, Suzi leverages the broader